

Education

University College London

Doctor of Philosophy, 2026.

PhD thesis: Layered monoidal theories.

Research group: Programming Principles, Logic and Verification.

Supervisor: Fabio Zanasi.

University of Amsterdam

MSc Logic (cum laude), 2020.

MSc Thesis: Quantum quirks, classical contexts: Towards a Bohrification of effect algebras.

Supervisors: Chris Heunen (University of Edinburgh), Nick Bezhanishvili.

University of Edinburgh

BSc Mathematics (First class honours), 2018.

BSc Project: Sheaves on topological spaces and their logic.

Supervisor: Tom Leinster.

Jyväskylän Normaalikoulu (Finland)

Ylioppilastutkinto (High school matriculation examination), 2014.

Mathematics, Physics, Chemistry, Finnish, Russian, English, French.

Work experience

University College London

Research fellow.

October 2025 - Present.

University College London

Various teaching assistant roles in mathematics and theoretical computer science.

January 2022 - December 2025.

Aboa Vetus Ars Nova museum

Civilian service year.

October 2020 - September 2021.

University of Amsterdam

Teaching assistant for MSc Mathematics course “Category Theory and Topos Theory”.

Autumn 2019.

Tartu Observatory

Satellite attitude determination software developer.

Summer 2018.

London Mathematical Society Undergraduate Research Bursary

Project title: Quantum logic and set theory.

Summer 2017.

Tartu Observatory

Erasmus+ traineeship.

Traineeship title: Development of Satellite Attitude Determination Software.

Summer 2016.

Department of Physics, University of Jyväskylä

Summer trainee, 2013.

Publications

Modulation-reaction networks

with Yoav Montacute.

Computational Methods in Systems Biology (To appear).

Preprint: arxiv.org/abs/2606.01193.

A categorical model for organic chemistry
with Ella Gale, Fabio Zanasi.
Theoretical Computer Science 1032 (2025). doi: 10.1016/j.tcs.2025.115084.

Learning closed signal flow graphs
with Ekaterina Piotrovskaya, Fabio Zanasi.
Theoretical Aspects of Computing – ICTAC (2024). doi: 10.1007/978-3-031-77019-7_5.

Disconnection rules are complete for chemical reactions
with Ella Gale, Fabio Zanasi.
Theoretical Aspects of Computing – ICTAC (2024). doi: 10.1007/978-3-031-77019-7_13.

A categorical approach to synthetic chemistry
with Ella Gale, Fabio Zanasi.
Theoretical Aspects of Computing – ICTAC (2023). doi: 10.1007/978-3-031-47963-2_17.

String diagrams for layered explanations
with Fabio Zanasi.
Electronic Proceedings in Theoretical Computer Science, 380 (2023). doi: 10.4204/EPTCS.380.21.

There and back again: A circuit extraction tale
with Miriam Backens, Hector Miller-Bakewell, Giovanni de Felice, John van de Wetering.
Quantum 5, 421 (2021). doi: 10.22331/q-2021-03-25-421.

Talks

International Conference on Computational Methods in Systems Biology 2026
Modulation-reaction networks.

1st Workshop on DIagrams in LLogic and Computation 2026
Towards a combinatorial representation of first-order bicategories.

Applied Category Theory 2026
Layered monoidal theories.

International Colloquium on Theoretical Aspects of Computing 2024
Disconnection rules are complete for chemical reactions.

Applied Category Theory 2024
Disconnection rules are complete for chemical reactions.

International Colloquium on Theoretical Aspects of Computing 2023
A categorical approach to synthetic chemistry.

Tenth Symposium on Compositional Structures 2022
Quantum quirks, classical contexts: Towards a Bohrification of effect algebras.

Applied Category Theory 2022
String diagrams for layered explanations.

Quantum Physics and Logic 2020
There and back again: A circuit extraction tale.

Refereeing

Journal referee

- International Journal of Quantum Information. worldscientific.com/worldscinet/ijqi.
- Compositionality. compositionality.episciences.org.
- Quantum. quantum-journal.org.

Conference referee (external)

- Applied Category Theory (ACT). 2026.
- International Conference on Formal Structures for Computation and Deduction (FSCD). 2026.
- European Symposium on Programming (ESOP). 2026.

- Theory of Quantum Computation, Communication and Cryptography (TQC). 2023.
- International Colloquium on Automata, Languages, and Programming (ICALP). 2023.
- Applied Category Theory (ACT). 2022.
- Quantum Physics and Logic (QPL). 2021.

Other service

Weekly seminar of the research group

Programming Principles, Logic and Verification, University College London.

Organiser, January – June 2026.

15th Meeting of the Southern and Midlands Logic Seminar

Local organiser, June 2026.

Languages

Finnish (native), Russian (native), English (fluent),
French (DEL F B2), Swedish (intermediate).